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LAB EXPERIMENT - 08

Q8. Execute and Display the Results of Nmap Service Version and OS Detection Commands in Kali Linux

AIM

To execute and display the results of the Nmap service version detection, aggressive scan, and operating system detection commands in the Kali Linux operating system and understand their basic functions.

PROCEDURE

1. Open the Kali Linux Terminal.
2. Verify that Nmap is installed using the nmap --version command.
3. Choose an authorized target (here, the local machine 127.0.0.1; scanme.nmap.org or a lab VM IP can also be used).
4. Execute the nmap -sV command to detect the versions of the services running on the open ports.
5. Execute the nmap -A command to perform an aggressive scan (OS detection, version detection, script scanning and traceroute).
6. Execute the sudo nmap -O command to detect the operating system of the target.
7. Observe and verify the output of all three commands and understand their functions.
8. Capture screenshots of the terminal showing the execution and output of the commands as evidence.

Note: Scan only systems you own or have written permission to test.

COMMANDS TO EXECUTE

First, check Nmap and set the target:

```
nmap --version
```

```
└─$ nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcre2-10.46 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select
```

Then execute:

1. Detect the Version of Services Running

The -sV option probes the open ports to determine the service name and version (for example, OpenSSH 6.6.1 or Apache httpd 2.4.7).

```
nmap -sV 127.0.0.1
```

OUTPUT:

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-13 12:12 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.39 seconds
```

2. Aggressive Scan

The -A option enables OS detection (-O), version detection (-sV), default script scanning (-sC) and traceroute in a single scan. It is noisy and easily detected, so use it only on authorized targets.

```
nmap -A 127.0.0.1
```

OUTPUT:

```
┌──(kali㉿kali)-[~]
└─$ nmap -A 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-13 12:13 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000069s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.84 seconds
```

3. Detect the Operating System of the Target

The -O option analyses TCP/IP stack responses to guess the operating system and its version. It requires root privileges, so run it with sudo.

```
sudo nmap -O 127.0.0.1
```

OUTPUT:



SUMMARY OF OPTIONS

Option	Function
-sV	Detects service names and version numbers on open ports
-A	Aggressive scan: OS detection, version detection, scripts and traceroute
-O	Detects the target operating system (needs root privileges)

RESULT

The Nmap service version detection (-sV), aggressive scan (-A) and operating system detection (-O) commands were successfully executed in Kali Linux. The -sV command displayed the versions of the services running on the open ports, the -A command performed a comprehensive scan of the target, and the -O command identified the operating system of the target.